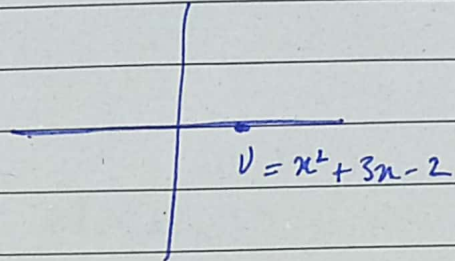


HOME WORK 3:-

PART A:-

Section 3.1 page: 131

Q71.

derivate wrt. to ~~time~~ x :-

$$v = x^2 + 3x - 2$$

$$v' = 2x + 3$$

for $x = 2$:-

$$v' = a = 2(2) + 3 = \boxed{7 \text{ cm/sec}^2}$$

$$Q74:- T = 2\pi \sqrt{\frac{L}{g}}$$

find

$$a) \frac{dT}{dL} :$$

$$\frac{dT}{dL} = \frac{2\pi}{\sqrt{g}} \cdot \frac{1}{2\sqrt{L}}$$

$$\frac{dT}{dL} = \frac{2\pi}{\sqrt{g}} \cdot \frac{1}{2\sqrt{L}}$$

$$\boxed{\frac{dT}{dL} = \frac{\pi}{\sqrt{g}\sqrt{L}}}$$

b) The sign of $\frac{dT}{dL}$ is positive, which shows a decrease in time as the length L increases.

Q75:- $A = \pi r^2$

a) find:- $\frac{dA}{dr} = 2\pi r$

b) Yes, this is the equation of circumference of circle.

c) $\frac{f(x) + f'(x)}{f'(x)} \quad \frac{f(r+h) - f(r)}{h}$

~~$\frac{\pi r^2}{r} = \pi r$~~

$$\lim_{h \rightarrow 0} \frac{\pi(r+h)^2 - \pi(r)^2}{h}$$

$$\lim_{h \rightarrow 0} \frac{\pi(r^2 + 2rh + h^2) - \pi r^2}{h}$$

$$\lim_{h \rightarrow 0} \frac{\cancel{\pi r^2} + 2\pi r h + h^2 - \cancel{\pi r^2}}{h}$$

$$\lim_{h \rightarrow 0} \frac{h[(2\pi r) + h]}{h}$$

$$\lim_{h \rightarrow 0} 2\pi r + h$$

apply limit-

$$= 2\pi r + 0 = \boxed{2\pi r}$$

Hence gives the circumference of circle.

PART B:-

Section 3.2:- page 135.

Q5:- $y = 5x^2 + 2^x + 3$

$$y' = 10x + 2^x \ln(2)$$

Q8:- $z = (\ln 4) e^x$

$$z' = (\ln 4) \cdot e^x$$

Q11:- $z = \ln(4) 4^x$

$$z' = \ln(4) \frac{dz}{dx} (4^x)$$

$$z' = \ln(4) \cdot 4^x \cdot \ln(4)$$

$$z' = [\ln(4)]^2 4^x$$

Q21. $f(t) = e^{t+2}$

$$f'(t) = e^{t+2}$$

$$\frac{df}{dt} = e^{t+2} (1) = e^{t+2}$$

Q25. $g(x) = 2x - \frac{1}{\sqrt[3]{x}} + 3^x - e$

$$g'(x) = 2 - x^{-1/3} + 3^x \ln(3) - e$$

$$g'(x) = 2 + \frac{1}{3} x^{-4/3} + 3^x \ln(3) - 0$$

$$g'(x) = \frac{1}{3x^{4/3}} + 3^x \ln(3) + 2 \quad \text{Ans.}$$

Q26:- $f(x) = (3x^2 + \pi)(e^x - 4)$

$$u \cdot v = u'v + v'u$$

$$u = 3x^2 + \pi \quad v = e^x - 4$$

$$u' = 6x \quad v' = e^x$$

$$6x(e^x - 4) + e^x(3x^2 + \pi)$$

$$f'(x) = 6xe^x - 24x + 3e^x x^2 + e^x \pi$$

Q22. $f(x) = e^{kx} - 1$

$$f'(x) = ke^{kx} - 0$$

$$f'(x) = ke^{kx}$$

Q27:- $f(x) = x^{\pi^2} + (\pi^2)^x$

$$f'(x) = \pi^2 x^{(\pi^2-1)} + \pi^{2x} \ln(\pi^2)$$

$$f'(x) = \pi^2 x^{(\pi^2-1)} + 2\pi^{2x} \ln(\pi)$$

Q38. $P(t) = 300 (1.044)^t$

$$P'(t) = 300 (1.044)^t \ln(1.044)$$

$$P'(5) = 300 (1.044)^5 \ln(1.044)$$

$$\boxed{P'(5) = 16.0}$$

Interpretation. Per year the number of animal rises by $\boxed{16}$.

Q39. $P = P_0 (1.05)^t$ ~~per year~~ early 5% inflation

$$P' = P_0 (1.05)^t \cdot \ln(1.05)$$

$$P = (1) (1.05)^{100} \cdot \ln(1.05)$$

$$P = 0.0794 \times 100 = \boxed{7.94 \%} \text{ cents/year}$$

Q41:- in 2009

Mexico:-

$$P_m = 111 \text{ million}$$

$$r_m = 1.13\% \text{ or } 0.0113$$

$$P_n = P_m r_m^{n \text{ yrs}} = \cancel{P_n = (111000000)($$

USA:-

$$P_u = 307 \text{ million}$$

$$r_u = 0.975\% \text{ or } 0.00975$$

$$P_n = P_u r_u^{n \text{ yrs}}$$

Solution:-

$$P_m = (111,000,000) (0.0113)^{n \text{ yrs}} + 1$$

$$P_u = (307,000,000) (0.00975)^{n \text{ yrs}} + 1$$

in 2009 $n=1$ for Mexico

$$\frac{dp}{dt} = (111,000,000) (1.0133)^n \ln(1.0133)$$

where $n=1$:-

$$\begin{aligned} \frac{dp}{dt} &= (111,000,000) (1.0133)^1 \ln(1.0133) \\ &= 1486074 \text{ people/year.} \end{aligned}$$

for USA:-

$$\frac{dp}{dt} = (307,000,000) (1.0097)^n \ln(1.0097)$$

where $n=1$ in 2009

$$\frac{dp}{dt} = (307,000,000) (1.0097)^1 \ln(1.0097)$$

$$\frac{dp}{dt} = 2992296 \text{ people/year.}$$

as $2992296 > 1486074$ people/year. Hence rate of increase of USA population is more than Mexican population. USA is growing faster.

Q43. a) $f(x) = 1 - e^x$

$$f'(x) = -e^x$$

$$\therefore f(x) = 0$$

$$0 = 1 - e^x$$

$$e^x = 1$$

$$x = \ln(1)$$

$$\boxed{x = 0}$$

$$f'(0) = -e^x$$

$$= -e^0$$

$$\boxed{f'(0) = -1} \text{ answer}$$

b) $m_1 m_2 = -1$

$$\boxed{m_2 = 1}$$

$$y - y_1 = m(x - x_1)$$

$$y - 0 = -1(x - 0)$$

$$\boxed{y = -x}$$

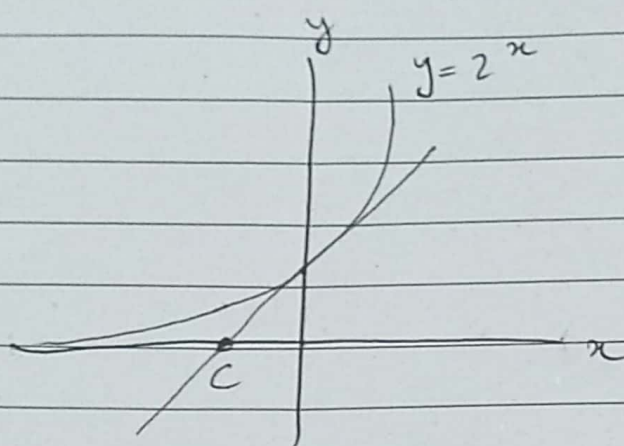
c) $m_2 = 1 \left[(0, 0) \right]$

$$y - y_1 = m(x - x_1)$$

$$y - 0 = 1(x - 0)$$

$$\boxed{y = x}$$

Q44.



$$y = 2^x$$

$$y' = 2^x \ln(2)$$

$$y' = \ln(2) 2^x$$

$$\boxed{y'(0) = \ln(2)}$$

$$m = \ln(2) \quad (0, 1)$$

$$y - y_1 = m(x - x_1)$$

$$y - 1 = x \ln(2)$$

$$x \ln 2 - y + 1 = 0$$

$$y = 0 \Rightarrow$$

$$x \ln(2) + 1 = 0 \quad \left(\frac{-1}{\ln(2)}, 0 \right)$$

$$x = \frac{-1}{\ln(2)}$$

$$\boxed{C = \left(\frac{-1}{\ln(2)}, 0 \right)} \text{ Ans.}$$